Explainable AI in Education: A Lightweight Web Dashboard for Student Academic Performance Prediction and Intervention

An Interpretable, SHAP-Driven Lightweight Framework for Transparent Student Risk Assessment and Pedagogical Intervention

Swapnil Kesare
Department of Artificial Intelligence and Data Science

Parth Katarmal
Department of Artificial Intelligence and Data Science

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Abstract:

Student academic-risk prediction can support early intervention, but practical educational deployment requires more than predictive accuracy. Many high-performing models are difficult for educators to interpret, while lightweight systems may provide limited explanations or actionable support. This manuscript presents a proposed lightweight and explainable framework for student academic performance prediction and intervention. The framework uses academic and behavioural indicators including attendance, assignment submission frequency, internal-test performance, and study hours. A preprocessing pipeline performs missing-value imputation, categorical encoding, numerical scaling, stratified train-test partitioning, and SMOTE-based class balancing on the training subset only. Four candidate classifiers—Decision Tree, Support Vector Classifier, Logistic Regression, and Random Forest—are compared using stratified k-fold cross-validation and grid-search tuning. SHAP is then used to expose local feature-level contributions for individual predictions. A lightweight Streamlit dashboard presents the predicted risk tier, explanation, and intervention recommendations, while a rule-based intervention layer maps dominant risk-increasing features to targeted pedagogical actions. The work is presented as a literature-grounded design study rather than a completed live experimental evaluation; therefore, no new accuracy claim is asserted. Reported studies indicate that lightweight educational classifiers can achieve competitive performance while reducing computational and interpretability barriers. The proposed framework combines prediction, explanation, and intervention in one educator-facing workflow intended for ordinary institutional hardware.

Keywords

Explainable AI; Educational Data Mining; Student Performance Prediction; SHAP; Early Warning System; Streamlit; Pedagogical Intervention

1. Introduction

Educational institutions increasingly use machine learning and educational data mining (EDM) to identify students who may require academic support. Early identification is valuable because an intervention can be more effective when it occurs before a failure becomes difficult to reverse. However, a practical educational decision-support system must also be understandable to teachers. A prediction such as “high risk” is less useful if the educator cannot determine which student characteristics produced that result [1].

The reviewed literature demonstrates a persistent trade-off between predictive performance, interpretability, and computational cost. Deep architectures such as graph neural networks and Bi-LSTM models can report strong predictive performance, but their training and deployment requirements may be difficult to support on ordinary institutional hardware. Lightweight models, on the other hand, can be easier to deploy but may not automatically provide a sufficiently detailed explanation of an individual prediction. The source methodology identifies this as the central research gap addressed by the proposed framework [1], [2], [3], [6].

This manuscript therefore proposes a five-phase framework that integrates data preprocessing, lightweight classification, SHAP-based explainability, an interactive Streamlit dashboard, and a rule-based intervention engine. The objective is not to claim superiority over deep learning, but to provide a transparent and accessible workflow in which prediction is directly connected to an understandable reason and an actionable response.

1.1 Research Gap

1. High-performing academic prediction models are often computationally complex and difficult for non-technical educators to interpret [2], [3], [6].
2. Many systems emphasize prediction but provide limited support for translating risk factors into concrete interventions [6], [7].
3. Static academic and demographic records can omit behavioural indicators and evolving learning patterns [8], [9], [12].
4. Educational institutions without specialized computing infrastructure require lightweight solutions with reasonable predictive capability [1], [14].
5. Binary risk outputs can overlook the practical value of distinguishing low, moderate, and high levels of academic risk [15].

1.2 Objectives

1. Develop a lightweight student academic-risk prediction pipeline using interpretable classical machine-learning classifiers.
2. Use SHAP to provide local feature-level explanations for individual risk predictions.
3. Provide an educator-facing Streamlit dashboard that presents prediction, explanation, and intervention information.

4. Map dominant risk-increasing factors to predefined pedagogical support actions.
5. Design the system for deployment on ordinary institutional hardware rather than specialized acceleration.

2. Literature Review and Related Work

The methodology report reviewed fifteen studies covering educational data mining, machine learning, deep learning, behavioural feature engineering, explainability, preprocessing, and hybrid prediction. The review found no single universally best-performing model and highlighted the recurring tension between accuracy and interpretability [1], [2], [6], [14]. Recent explainable deep-learning systems demonstrate high reported accuracy but introduce computational and structural complexity [2], [3]. Lightweight classifiers and ensembles offer a more practical deployment path, particularly where institutional hardware and technical support are limited [1], [6], [14].

The literature also indicates that feature selection matters. Motivation, time-management self-efficacy, attendance, assignment activity, internal assessments, study behaviour, and learning-management-system telemetry can contribute useful signals [8], [9], [12]. Data preprocessing, missing-value handling, normalization, feature selection, and class-imbalance management are important for improving generalization [1], [13], [14]. The reviewed studies further suggest that continuous behavioural indicators can be more informative than static registration information alone [9], [12].

Table 1 summarizes the reviewed evidence and the gap relevant to the proposed framework.

No.	Study	Methodology	Reported result	Gap / implication
1	Pan et al. (2025)	Survey of ML/EDM	No consensus on best model	Need practical model selection
2	Jha et al. (2025)	XGNN-AP + SHAP/GNN Explainer	99.71% accuracy; F1=1.00	High complexity
3	Kalita et al. (2025)	Bi-LSTM + SHAP	88.23% accuracy	High compute / opaque network
4	Angeioplastis et al. (2025)	kNN/NN/LMS EDM	F1 > 0.80	Ignores dynamics
5	Gonzales et al. (2025)	Hybrid ML-DL	Higher sensitivity	Few basic-school models
6	Johora & Hasan (2025)	Voting ensemble + SHAP/LIME	86.38% accuracy	Trust barrier
7	Paul et al. (2025)	Regression + XAI	Forecasted first-semester GPA	Late detection
8	Lu et al. (2024)	ML + statistical analysis	Motivation impacts GPA	Behavioural traits overlooked
9	Luo et al. (2024)	Spatio-temporal EDM	Higher trajectory precision	Static-data dependence
10	Pires et al. (2024)	Supervised ML for programming	Forecasted coding risk	Needs domain tailoring

11	Hussain et al. (2024)	Decision tree / rule mining	Transparent pathways	Lacks accessible pipeline
12	Liu et al. (2023)	Clickstream + ML	Logs linked to exam marks	Underused LMS telemetry
13	Feng et al. (2022)	Data cleaning / feature selection	Reduced noise and overfitting	Data curation issues
14	Zhang et al. (2021)	EDM/ML/DL comparison	Accuracy–compute trade-off	Hardware limitations
15	Alshanqiti & Namoun (2020)	Multi-label regression	Granular grade tiers	Binary outcome is coarse

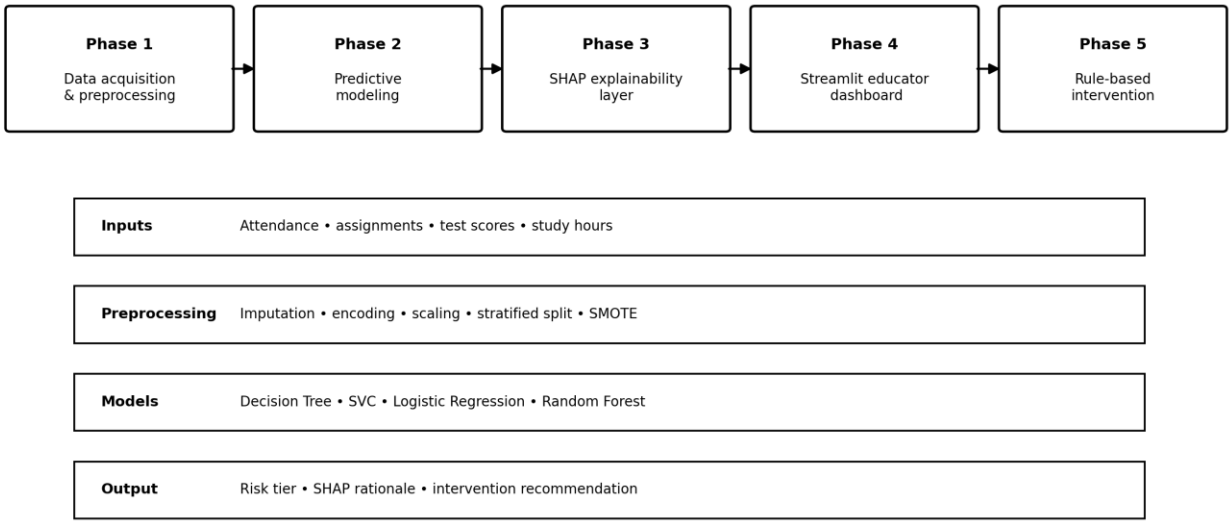
2.1 Positioning of the Proposed Work

The proposed system occupies the lightweight, educator-facing end of the design space. It intentionally combines classical classifiers with SHAP rather than replacing them with a computationally heavier deep architecture. Its distinctive emphasis is the closed loop from prediction to explanation to intervention: the system is intended to help an educator understand why a student is flagged and what support action may be appropriate [1]–[3], [6].

3. Methodology / Proposed Method

The proposed framework follows five phases: (1) data acquisition and preprocessing, (2) predictive modelling, (3) SHAP explainability, (4) interactive web deployment, and (5) rule-based intervention. The source methodology specifies academic and behavioural inputs such as attendance, assignment submissions, internal test scores, and study hours, followed by preprocessing and stratified partitioning. SMOTE is applied only to the training subset to address class imbalance while protecting the evaluation data from synthetic samples [1], [13].

Figure 1. Proposed Five-Phase Explainable Student-Risk Prediction Workflow



3.1 Data Acquisition and Preprocessing

The input layer contains routine student indicators that can be collected during a term. The core feature families specified in the methodology are attendance rate, assignment submission frequency, internal-test scores, and study hours. Data preparation consists of missing-value imputation, categorical encoding where required, and numerical feature scaling. The data are then divided into training and testing subsets using stratified partitioning. SMOTE is restricted to the training data so that the independent evaluation set retains its original distribution. This separation is an important anti-leakage measure [1], [13].

ID	Feature	Category	Description	Expected risk relation
F1	Attendance rate	Academic / behavioural	Percentage of attended classes	Lower attendance may increase risk
F2	Assignment submission frequency	Academic behaviour	Frequency or proportion of submitted work	Lower submission may increase risk
F3	Internal-test score	Academic performance	Internal assessment performance	Lower score may increase risk
F4	Study hours	Behavioural	Reported study time	Lower study time may increase risk

3.2 Predictive Modelling

Four candidate classifiers are trained: Decision Tree, Support Vector Classifier (SVC), Logistic Regression, and Random Forest. They represent complementary modelling assumptions. Decision Trees provide transparent rule-like splits; Logistic Regression provides a simple linear probabilistic baseline; SVC provides a margin-based classification approach; and Random Forest provides ensemble-based robustness. Stratified k-fold cross-validation and grid search are used for model comparison and hyperparameter tuning [1], [13].

Model / module	**Role**	**Main tuning parameters**	**Interpretability**
Decision Tree	Transparent baseline	max_depth; min_samples_split; criterion	High — rule-based splits
SVC	Margin-based classifier	C; kernel; gamma	Low — SHAP required
Logistic Regression	Linear probabilistic baseline	C; penalty; solver	High — coefficients readable
Random Forest	Ensemble classifier	n_estimators; max_depth; min_samples_leaf	Moderate — importances + SHAP
SHAP	Local/global explanation layer	Background sample; explainer type	Primary interpretability mechanism

3.3 Explainable AI Layer

SHAP (Shapley Additive Explanations) is applied after model training to estimate the contribution of each input feature to an individual prediction. For a student classified as at risk, the explanation can show which variables pushed the prediction toward the risk class and which variables pushed it away. The source methodology proposes waterfall and force-style visualizations for communicating local explanations to educators. This design makes explainability operational rather than purely diagnostic. Instead of presenting only a probability, the dashboard is intended to expose the dominant contributors so that the educator can decide whether an intervention is appropriate [2], [3], [6], [7].

3.4 Risk-Tier Assignment

The proposed inference logic converts the predicted probability into three risk tiers. These thresholds are defined in the source methodology as high risk for $p \geq 0.66$, moderate risk for $0.33 \leq p < 0.66$, and low risk for $p < 0.33$. These thresholds are design parameters and should be recalibrated against an institution's validated data before high-stakes use.

Figure 4. Risk-Tier Decision Thresholds Used in the Proposed Inference Logic

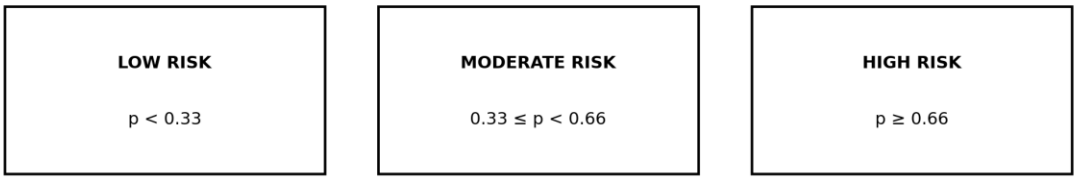


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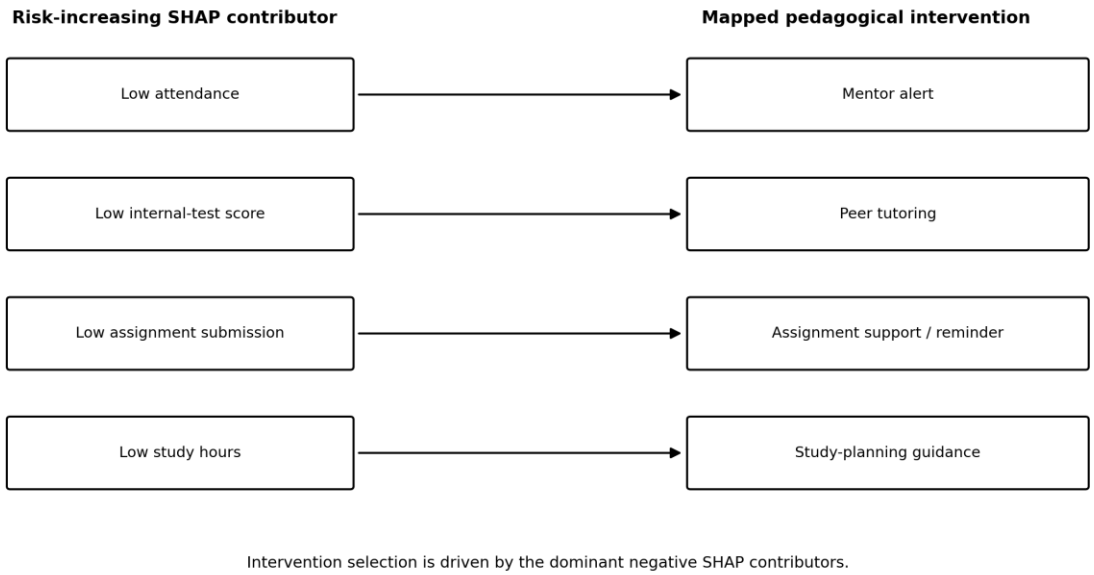
3.5 Interactive Dashboard

The educator-facing interface is implemented conceptually as a lightweight Streamlit dashboard. Educators enter routine student metrics through form inputs or sliders and receive the predicted risk category, SHAP explanation, and recommended intervention. This separates the machine-learning implementation from the user experience and reduces the technical knowledge required to operate the system.

3.6 Intervention Engine

The intervention layer inspects dominant negative SHAP contributors and maps them to predefined support actions. The source methodology gives four examples: low attendance to a mentor alert, low internal-test score to peer tutoring, low assignment submission to assignment support or reminders, and low study hours to study-planning guidance. The mapping is intentionally rule-based so that educators can inspect and modify the intervention logic.

Figure 3. Rule-Based Intervention Mapping



Intervention selection is driven by the dominant negative SHAP contributors.

Figure 3. Rule-Based Intervention Mapping

3.7 System Architecture

Figure 2 presents the component-level architecture. Raw student records are transformed by the preprocessing pipeline, the balanced training data are used to obtain a tuned classifier, and the inference engine combines classification with SHAP explanations. The dashboard presents the resulting risk and rationale, while the intervention engine converts dominant risk contributors into support recommendations.

Figure 2. System Component Architecture

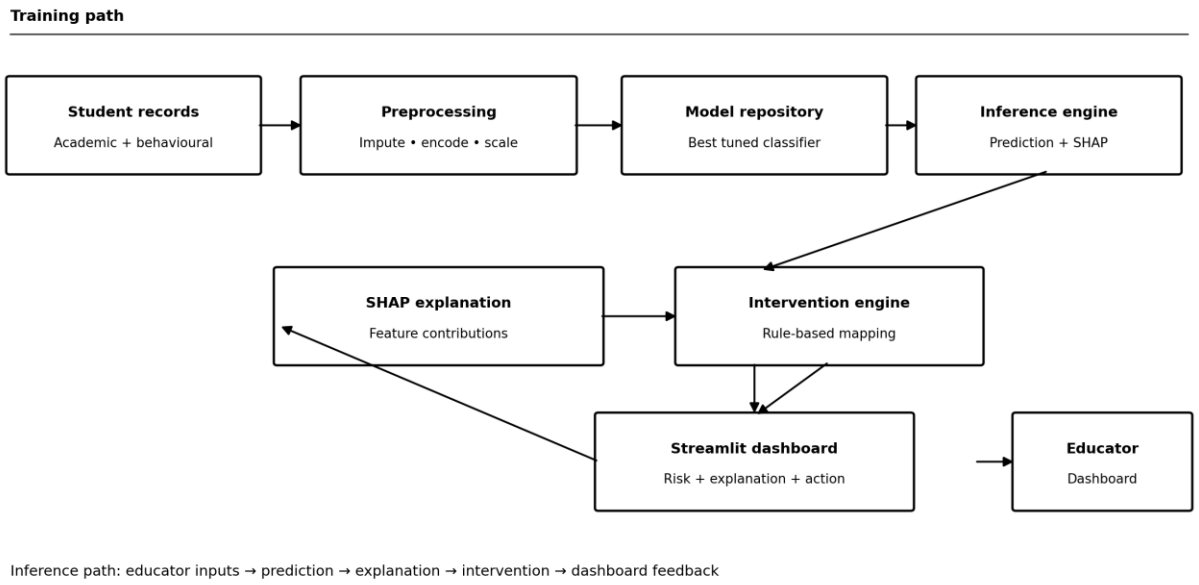


Figure 2. System Component Architecture

3.8 Algorithmic Procedure

Algorithm 1 formalizes the proposed inference process. The procedure preprocesses the student vector, computes the risk probability, assigns a risk tier, calculates local SHAP values, ranks risk-increasing features, maps them to interventions, and renders the result on the dashboard. It also allows decision information to be logged for longitudinal monitoring subject to privacy requirements.

1. Input: student feature vector containing attendance, assignment submission frequency, internal-test score, and study hours.

2. Apply the fitted preprocessing pipeline learned from the training data.

3. Compute predicted probability p using the selected classifier.

4. Assign risk tier: High if $p \geq 0.66$; Moderate if $0.33 \leq p < 0.66$; Low if $p < 0.33$.

5. Compute local SHAP attributions using a compatible explainer and training-data background distribution.

6. Rank features according to their contribution toward the at-risk/high-risk prediction.

7. Map dominant risk-increasing features to the predefined intervention rule base.

8. Render the risk tier, SHAP explanation, and intervention recommendations on the Streamlit dashboard.

9. Log appropriate prediction/explanation information for monitoring, subject to privacy requirements.

10. Return the risk tier, SHAP attribution vector, and intervention set.

4. Tools, Technologies and Experimental Setup

The implementation is designed around a Python 3.10+ software stack. NumPy and pandas support numerical and tabular processing, scikit-learn provides preprocessing and classical classifiers, imbalanced-learn provides SMOTE, SHAP provides feature-level explanations, and Streamlit provides the educator-facing dashboard. Matplotlib, Seaborn, and Plotly are identified in the methodology as visualization libraries. The intended hardware profile is an ordinary multi-core CPU with 8–16 GB RAM and an SSD; a GPU is optional rather than required.

Category	**Component**	**Purpose**
Operating system	64-bit Windows 11 / Linux Ubuntu 22.04 LTS	Development and execution
Programming language	Python 3.10+	Core implementation
Data processing	pandas, NumPy, scikit-learn	Cleaning, encoding, scaling, modelling
Class imbalance	imbalanced-learn (SMOTE)	Training-set oversampling
Models	Decision Tree, SVC, Logistic Regression, Random Forest	Risk-tier classification
Explainability	SHAP; LIME listed in methodology	Feature attribution / XAI
Dashboard	Streamlit	Interactive educator interface
Visualization	Matplotlib, Seaborn, Plotly	Evaluation and explanation charts
Development	Jupyter Notebook	Experimentation and validation

Component	**Specification**	**Purpose**
Processor	Intel Core i5 / AMD Ryzen 5 multi-core	Classifier training and SHAP computation
RAM	8–16 GB DDR4	SMOTE and tabular processing
Storage	256 GB SSD	Dataset and model I/O
GPU	Not required; optional for large-scale tuning	Optional acceleration

5. Results and Discussion

5.1 Nature of the Evaluation

The present manuscript is a literature-grounded design study based on the methodology report. The supplied source does not report a completed live experimental run on a specified institutional dataset. Consequently, this manuscript does not invent a new accuracy, precision, recall, F1-score, ROC-AUC, or confusion matrix for the proposed system. Instead, reported values from the reviewed studies are used as contextual benchmarks, and the proposed 85–90% accuracy range is treated only as a preliminary benchmark for future experimental evaluation, not as a result achieved by this work.

Approach	Reported metric	Model family	Interpretation
XGNN-AP + SHAP/GNN Explainer	99.71%	Deep / graph-based	High reported performance; higher complexity
Bi-LSTM + SHAP	88.23%	Deep / temporal	Captures temporal dependencies; computationally heavier
Soft-voting ensemble + SHAP/LIME	86.38%	Classical ensemble	Explainable and comparatively lightweight
EDM classifiers	F1 > 0.80	Classical ML	Competitive classical baseline
Proposed framework	Not experimentally established	Lightweight + SHAP	Designed for transparent educator-facing deployment

5.2 Discussion

The literature indicates that deep architectures can deliver strong predictive performance, but their complexity may reduce accessibility for institutions with limited computing infrastructure. The proposed framework therefore makes an explicit design trade-off: it prioritizes lightweight classifiers and transparent explanations rather than attempting to outperform the most complex deep-learning systems. This is consistent with the source conclusion that the principal contribution is the integration of prediction, explanation, and intervention within one workflow[1], [6], [7].

A second methodological strength is the placement of SMOTE only after the training-test split. Applying synthetic oversampling to the full dataset before partitioning could contaminate the evaluation set and lead to overly optimistic estimates. Restricting SMOTE to training data preserves a cleaner evaluation design. The framework also benefits from stratified cross-validation because the risk classes may not be equally represented [2], [3], [6].

Finally, the intervention layer changes the role of explainability. A SHAP explanation is not treated only as a visual justification of a prediction; it becomes an input to a controlled rule base that proposes support

actions. This creates a direct chain from evidence to educational response, while retaining the possibility for human review [1], [6], [14].

6. Practical / Application Analysis

The framework is intended for practical early-warning use by educators, academic advisors, or institutional support teams. A typical operating scenario begins with routine student records being entered or imported into the system. The model estimates a risk tier, SHAP identifies the strongest contributors, and the intervention engine presents candidate actions. The educator can then review the recommendation in context rather than treating the model output as an automatic decision.

Operating scenario	System input/output	Practical purpose
Early-semester screening	Routine attendance, assignments, tests and study hours	Identify students needing attention before failure
Teacher/advisor review	Risk tier + SHAP explanation	Understand the principal factors behind a flag
Targeted support	Rule-mapped intervention	Select tutoring, reminders, mentoring or study-planning support
Longitudinal monitoring	Logged predictions where permitted	Track changes and refine intervention rules

6.1 Practical Constraints

1. The quality of predictions depends on the completeness, reliability, and representativeness of student data.
2. Risk thresholds should be calibrated and validated before use in consequential academic decisions.
3. SHAP explanations describe model behaviour; they do not prove that a feature is a causal reason for poor performance.
4. Intervention recommendations should support—not replace—teacher judgement and student-specific context.
5. Privacy, access control, data minimization, and appropriate institutional governance are required when handling student records.

7. Limitations and Future Scope

The principal limitation of the present work is that the supplied methodology is a proposed framework rather than a completed empirical study. A real deployment therefore requires a validated dataset, reproducible training runs, and quantitative evaluation on an independent test set. The literature-based benchmark cannot substitute for these experiments.

1. Evaluate the framework on a sufficiently large and representative institutional dataset.

2. Report accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, calibration, and class-specific performance.
3. Compare tuned classifiers using identical splits and preprocessing pipelines.
4. Perform ablation studies to quantify the effect of SMOTE, behavioural features, SHAP, and intervention mapping.
5. Evaluate explanation usefulness with educators rather than relying only on model-level explainability metrics.
6. Extend the feature space with validated LMS clickstream, temporal engagement, and domain-specific indicators where available.
7. Investigate fairness and subgroup performance to reduce the risk of systematically disadvantaging particular student groups.
8. Add model monitoring and threshold recalibration as academic patterns and cohorts change over time.
9. Integrate secure institutional data pipelines and role-based access controls for real-world deployment.

8. Conclusion

This manuscript presents a lightweight, explainable framework for student academic-risk prediction and intervention. The design responds to a recurring limitation in the literature: highly accurate models may be difficult to deploy and interpret, while simpler systems may not provide sufficiently actionable explanations. The proposed solution combines preprocessing and SMOTE-based imbalance handling, Decision Tree/SVC/Logistic Regression/Random Forest modelling, SHAP-based local explanations, a Streamlit dashboard, and rule-based intervention mapping. The resulting workflow is intended to provide educators with not only a risk prediction but also a human-readable rationale and a possible support action. The source methodology concludes that this combination offers a practical balance among predictive performance, transparency, computational accessibility, and intervention support.

9. Declarations

Declaration	**Statement**
Funding	No external funding is reported in the supplied methodology.
Conflict of interest	No conflict of interest is reported in the supplied methodology.
Author contribution	**Swapnil Kesare:** Conceptualization, methodology review, framework design, machine-learning pipeline design, preprocessing and predictive-modelling workflow, SHAP-based explainability design, dashboard and intervention workflow design, manuscript drafting, revision, and finalization. **Parth Katarmal:** conceptualization, methodology, literature synthesis, system design, and manuscript preparation.

Data availability	No original institutional dataset is reported in the supplied methodology; future empirical evaluation should document dataset access and governance.
Ethics and privacy	Any real-world deployment should comply with institutional requirements for student-data privacy, access control, and responsible use.

10. References

[1] J. Pan, Z. Zhao, and D. Han, "Academic performance prediction using machine learning approaches: A survey," IEEE Trans. Learn. Technol., vol. 18, pp. 351–368, 2025, doi: 10.1109/TLT.2025.3554174.

[2] M. K. Jha, K. Kumar, N. Hemrajani, D. S. Rao, A. Goyal, and R. Ajmera, "AI powered student performance prediction using explainable ML," in Proc. 4th Int. Conf. Automation, Computing and Renewable Systems (ICACRS), 2025, pp. 1140–1144, doi: 10.1109/ICACRS67045.2025.11324137.

[3] E. Kalita, A. M. Alfarwan, H. El Aouifi, A. Kukkar, S. Hussain, T. Ali, and S. Gaftandzhieva, "Predicting student academic performance using Bi-LSTM: A deep learning framework with SHAP-based interpretability and statistical validation," Front. Educ., vol. 10, Art. no. 1581247, 2025, doi: 10.3389/feduc.2025.1581247.

[4] A. Angeioplastis, J. Aliprantis, M. Konstantakis, and A. Tsimpiris, "Predicting student performance and enhancing learning outcomes: A data-driven approach using educational data mining techniques," Computers, vol. 14, no. 3, Art. no. 83, 2025, doi: 10.3390/computers14030083.

[5] W. Gonzales, Z. Cordero, C. D. Abanto-Ramírez, E. T. Susanibar Ramírez, H. Iftikhar, and J. L. López-Gonzales, "A hybrid AI approach for predicting academic performance in RBE students," Front. Artif. Intell., vol. 8, Art. no. 1651100, 2025, doi: 10.3389/frai.2025.1651100.

[6] F. T. Johora and M. N. Hasan, "An explainable AI-based approach for predicting undergraduate students academic performance," Array, vol. 26, Art. no. 100384, 2025, doi: 10.1016/j.array.2025.100384.

[7] R. Paul, S. Sarker, H. El Aouifi, S. Hussain, A. K. Baruah, and S. Gaftandzhieva, "Analyzing dropout of students and an explainable prediction of academic performance utilizing artificial intelligence techniques," Front. Educ., vol. 10, Art. no. 1698505, 2025, doi: 10.3389/feduc.2025.1698505.

[8] J. Lu, Y. Liu, S. Liu, Z. Yan, X. Zhao, Y. Zhang, C. Yang, H. Zhang, W. Su, and P. Zhao, "Machine learning analysis of factors affecting college students' academic performance," Front. Psychol., vol. 15, Art. no. 1447825, 2024.

[9] Z. Luo, J. Mai, C. Feng, D. Kong, J. Liu, Y. Ding, B. Qi, and Z. Zhu, "A method for prediction and analysis of student performance that combines multi-dimensional features of time and space," Mathematics, vol. 12, no. 22, Art. no. 3597, 2024.

[10] J. P. J. Pires, F. Brito Correia, A. Gomes, A. R. Borges, and J. Bernardino, "Predicting student performance in introductory programming courses," Computers, vol. 13, no. 9, Art. no. 219, 2024.

[11] M. M. Hussain, S. Akbar, S. A. Hassan, M. W. Aziz, and F. Urooj, “Prediction of student’s academic performance through data mining approach,” Journal of Informatics and Web Engineering, vol. 3, no. 1, pp. 241–251, 2024.

[12] Y. Liu et al., “Predicting student performance using clickstream data and machine learning,” Educ. Sci., vol. 13, no. 1, Art. no. 17, 2023.

[13] G. Feng, M. Fan, and Y. Chen, “Analysis and prediction of students’ academic performance based on educational data mining,” IEEE Access, vol. 10, pp. 19558–19571, 2022, doi: 10.1109/ACCESS.2022.3151652.

[14] Y. Zhang, Y. Yun, R. An, J. Cui, H. Dai, and X. Shang, “Educational data mining techniques for student performance prediction: Method review and comparison analysis,” Front. Psychol., vol. 12, Art. no. 698490, 2021.

[15] A. Alshanqiti and A. Namoun, “Predicting student performance and its influential factors using hybrid regression and multi-label classification,” IEEE Access, vol. 8, pp. 203827–203844, 2020, doi: 10.1109/ACCESS.2020.3036572.